

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

sns.set(style="whitegrid")
%matplotlib inline

import urllib.request

# URL to the Titanic dataset
url = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'

# Download and save as 'titanic.csv'
urllib.request.urlretrieve(url, 'titanic.csv')

# Load into a DataFrame
import pandas as pd
df = pd.read_csv('titanic.csv')

# Preview the data
df.head()
```



	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 2117
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 1759
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2 310128

Next steps:

[Generate code with df](#)

 [View recommended plots](#)

[New interactive sheet](#)

```
# Dataset info
df.info()

# Summary statistics
df.describe()

# Missing values
df.isnull().sum()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
```

```

10 Cabin          204 non-null    object
11 Embarked       889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

```

0
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2

```

```
dtype: int64
```

```

import seaborn as sns
import matplotlib.pyplot as plt

```

```

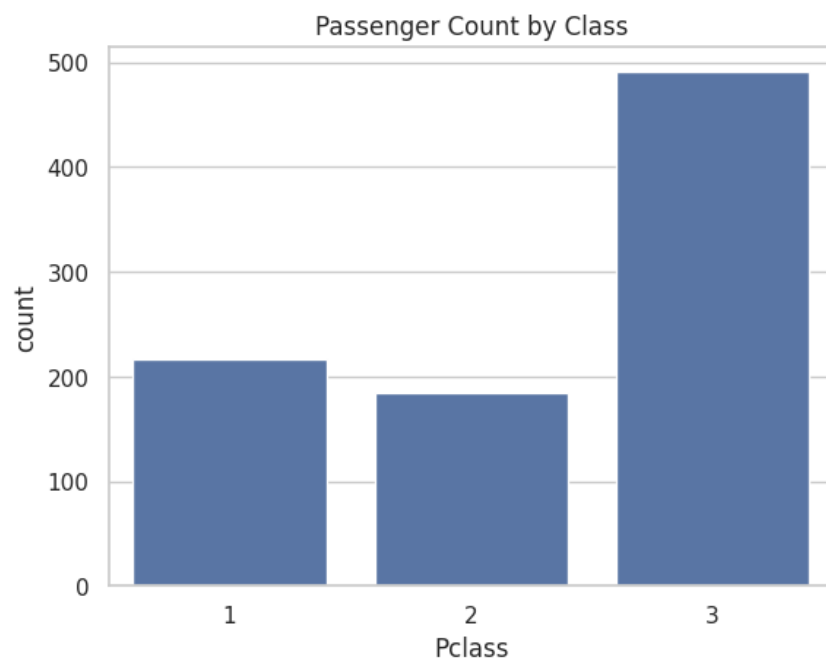
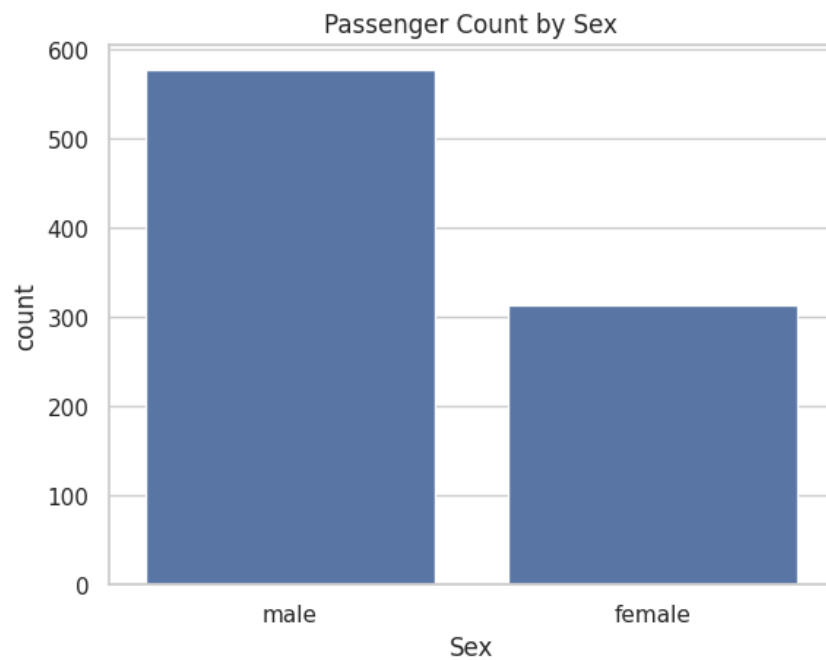
# Count of passengers by Sex
sns.countplot(x='Sex', data=df)
plt.title("Passenger Count by Sex")
plt.show()

```

```

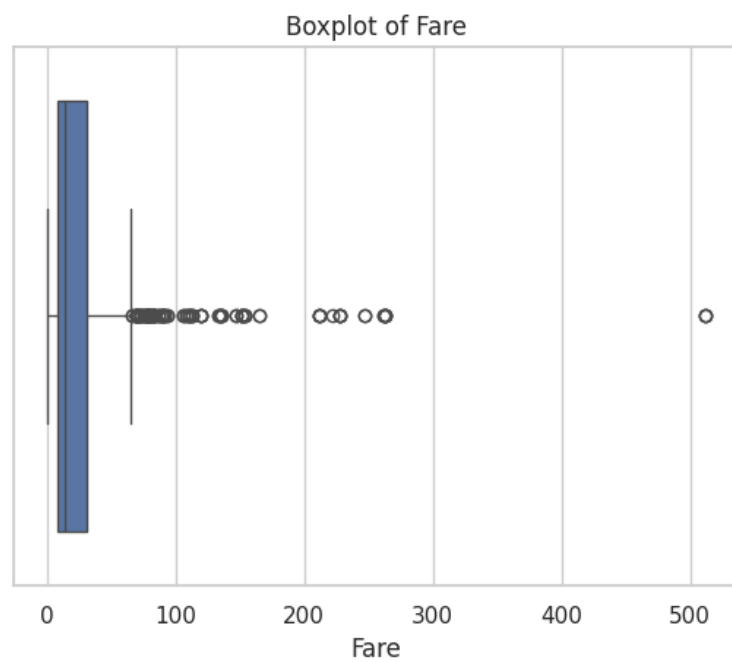
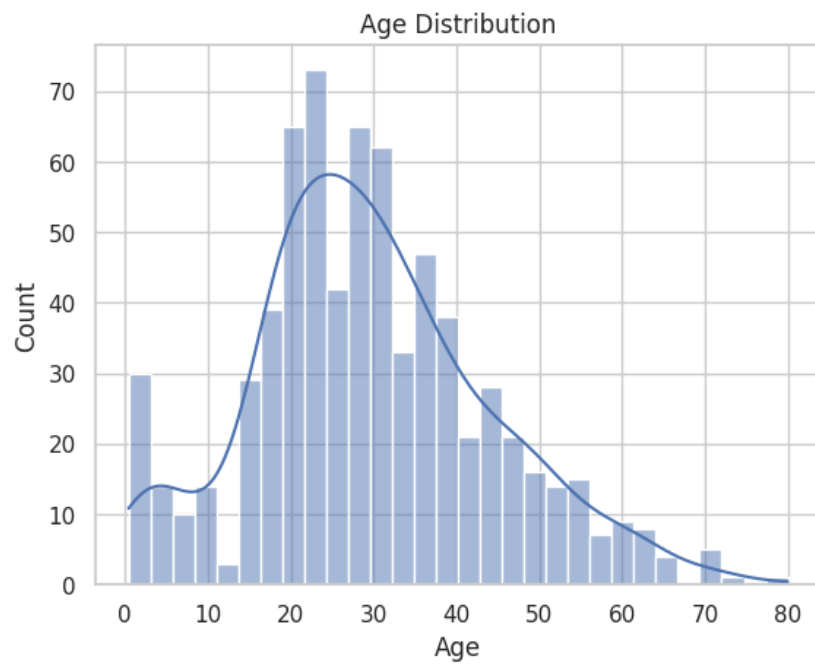
# Count of passengers by Class
sns.countplot(x='Pclass', data=df)
plt.title("Passenger Count by Class")
plt.show()

```



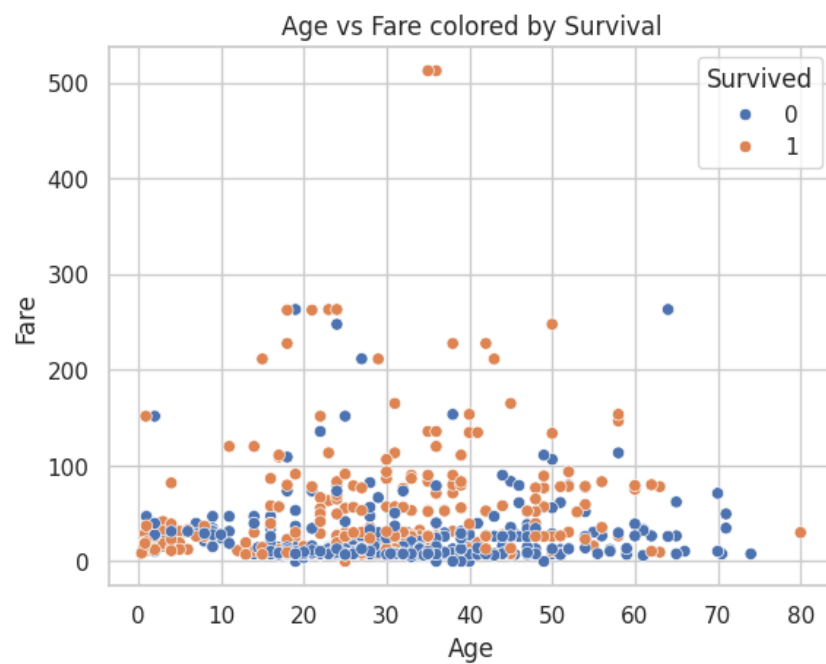
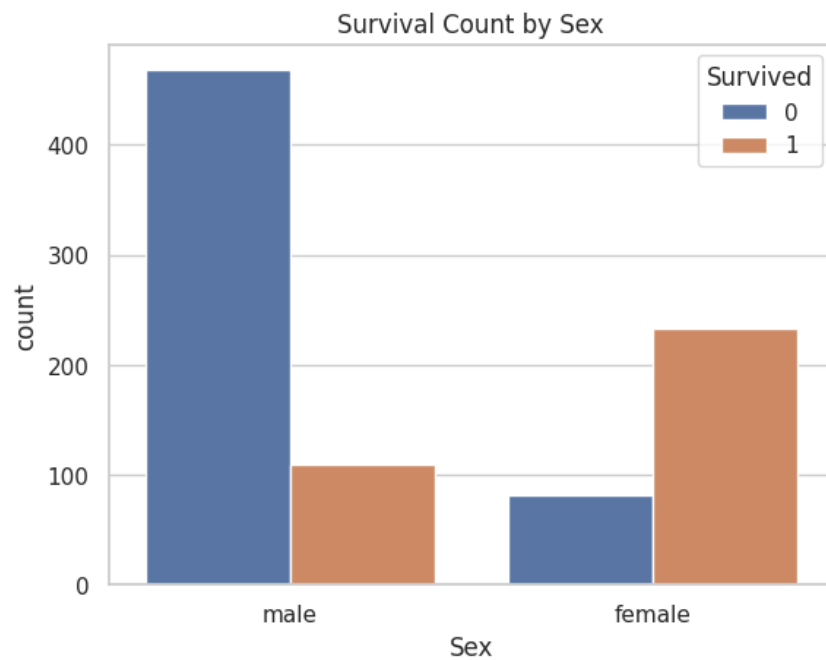
```
# Age Distribution
sns.histplot(df['Age'].dropna(), kde=True, bins=30)
plt.title("Age Distribution")
plt.show()
```

```
# Fare Distribution
sns.boxplot(x=df['Fare'])
plt.title("Boxplot of Fare")
plt.show()
```



```
# Survival rate by Sex
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival Count by Sex")
plt.show()

# Age vs Fare
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare colored by Survival")
plt.show()
```



```
# Select only numeric columns
numeric_df = df.select_dtypes(include=['number'])

# Correlation heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```

